

Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/radcr

Case Report

Case report of a very rare primary malignant bone tumor: Mixed type maxillary osteosarcoma ☆☆☆

Eko Ardianto Margiono, MD*, Sri Andreani, MD

Division of Neuro Radiology and Head & Neck Radiology, Department of Radiology, Faculty of Medicine, Universitas Airlangga–Dr Soetomo Academic General Hospital, Jalan Mayjen. Prof Dr Moestopo 47, Surabaya 60131, Indonesia

ARTICLE INFO

Article history:

Received 27 December 2022

Revised 27 January 2023

Accepted 29 January 2023

Keywords:

Osteosarcoma

Malignant bone tumor

Maxillary sinus

ABSTRACT

Osteosarcoma (OS) is the most common primary malignant bone tumor, accounting for approximately 20% of all primary bone malignancies. OS affects 2–4.8 in 1,000,000 individuals annually and is more common in men than women (1.5:1 ratio). The most common locations include the femur (42%), the tibia (19%), and the humerus (10%), whereas other potential sites include the skull or jaw (8%) and the pelvis (8%). We report a very rare case in a 48-year-old female who presented with swelling of the left cheek accompanied by palpable solid mass, with the final diagnosis of mixed type maxillary OS was confirmed from a surgical biopsy.

© 2023 The Authors. Published by Elsevier Inc. on behalf of University of Washington.

This is an open access article under the CC BY-NC-ND license

(<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Introduction

Osteosarcoma (OS) is the most common primary malignant bone tumor, accounting for approximately 20% of all primary bone malignancies. OS affects 2–4.8 in 1,000,000 individuals annually and is more common in men than women (1.5:1 ratio). The etiology of OS remains unclear, but possible risk factors include trauma, exposure to chemicals and carcinogens, and viruses. OS is characterized by the production of osteoid (immature bone tissue) by malignant mesenchymal cells. According to the World Health Organization's histologic classification of bone tumors, OS can be divided into central, intramedullary, and surface tumors, with a number of subtypes within each group. A diagnosis of OS can be made based on

the patient's history, a physical examination, radiography, and histopathology.

Case description

A 48-year-old female patient presented with swelling of the left cheek. Initially 3 years ago presented a palpable mass was as large as the tip of the fifth finger on the left cheek without any pain. For 3 years, patient did not carry out further examination or any treatment. In the following 3 year the mass grew larger to the size of an adult's fist a thumb and accompanied by pain. At this point, the patient also presented with left epiphora, exophthalmos, and blurry vision; rhinitis; and had

☆ Funding: None.

☆☆ Competing Interests: The authors declare no conflicts of interest.

* Corresponding author.

E-mail address: ekoardiantomargiono@yahoo.com (E.A. Margiono).<https://doi.org/10.1016/j.radcr.2023.01.104>1930-0433/© 2023 The Authors. Published by Elsevier Inc. on behalf of University of Washington. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

lost 23 kilograms in 6 months. The patient was still able to eat and drink, and there were no complaints of headaches or seizures. No palpable lymph nodes were present in the neck or other part of the body. The patient did not have a personal history of diabetes, hypertension, or any malignancy. History of previous treatment of chemotherapy, radiotherapy or surgery was denied.

A physical examination revealed a single mass 6 cm × 5 cm × 10 cm in the left maxillary-nasal-infraorbital region, fixed, hard-dense consistency, irregular surface, indistinct boundaries, and pain. The ulcerated mass was palpable intraorally in the left maxillary gingiva. On visual examination, the patient visus oculus dexter and visus oculus sinister were 5/5 and 5/6, respectively, and intraocular pressure was normal.

Panoramic radiographs showed permeative type maxillary sinus destruction with a sunburst periosteal reaction, a broad transitional zone, and chondroid type matrix. From the anamnesis, physical examination, and panoramic radiographs, the patient was diagnosed early with a differential diagnosis of OS, chondrosarcoma, and sarcoma. Then a computed tomography (CT) scan and fine-needle aspiration biopsy (FNAB) examinations were carried out. CT suggested the presence of an aggressive primary malignant bone tumor (most likely chondrogenic type OS); right and left maxillary sinusitis; right submandibular lymphadenopathy; and subcentimetric left submandibular, submental, and bilateral posterior trigonum

lymph nodes. There were no pulmonary metastases observed on thoracic radiography. A FNAB revealed hypercellular specimens with anaplastic cells with oval-round pleomorphic nuclei. Some cells were binucleated and hyperchromatic, with decent cytoplasm and eosinophilic matrix. The case was carried out in a multidisciplinary discussion with head and neck surgery, oral surgery, ophthalmology, radiology and pathology clinic with FNAB and CT scans previously carried out. From the multidisciplinary discussion the team has some conclusions by being diagnosed with OS regio maxillaris sinistra T3N0M0 with complications of exophthalmus OS and obstruction of the nasolacrimal duct OS. Then the patient underwent a total maxillectomy sinistra, extended exenteration OS, and reconstruction with titanium mesh. From the operation, an anatomic pathologic examination was carried out on the removed tissue with the results of a pathologic study revealed that the tumor was of mixed OS, chondroblastic, osteoblastic, and fibroblastic type. So that the final result of the diagnosis is mixed type maxillary OS.

Discussion

OS is a rare malignant osseous neoplasm. It arises near the metaphyses of long bones of the appendicular skeleton. The most common locations include the femur (42%), the tibia



Fig. 1 – Clinical presentation before surgery.



Fig. 2 – Panoramic evaluation: permeative type maxillary sinus destruction, sunburst periosteal reaction, a broad transitional zone, and chondroid type matrix.

(19%), and the humerus (10%), whereas other potential sites include the skull or jaw (8%) and the pelvis (8%) [1]. The incidence of OS is 3.4 cases per million people per year, with 8–11 cases per million people per year among individuals 15–19 years of age. Historically, the reported incidence is also higher in males [2,3]. Individuals of black race are most likely to be affected by OS, with an incidence rate of 6.8 cases per million people per year. Individuals of Hispanic descent are a close second with an incidence rate of 6.5 cases per million people per year, whereas the incidence rate in Caucasians is 4.6 cases per million people per year [4].

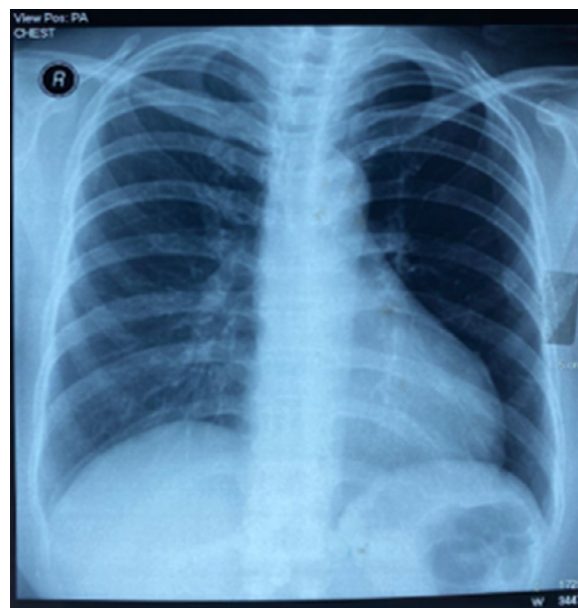


Fig. 4 – Radiological thoracic evaluation: no evidence of pulmonary metastasis.

The overall survival rate for patients with OS is 10%–20%. However, in the 1970s, the introduction of adjuvant chemotherapy for OS increased survival rates dramatically to 60%–70% for patients without metastases and 20%–30% for patients with recurrent or metastatic diseases [2]. The risk

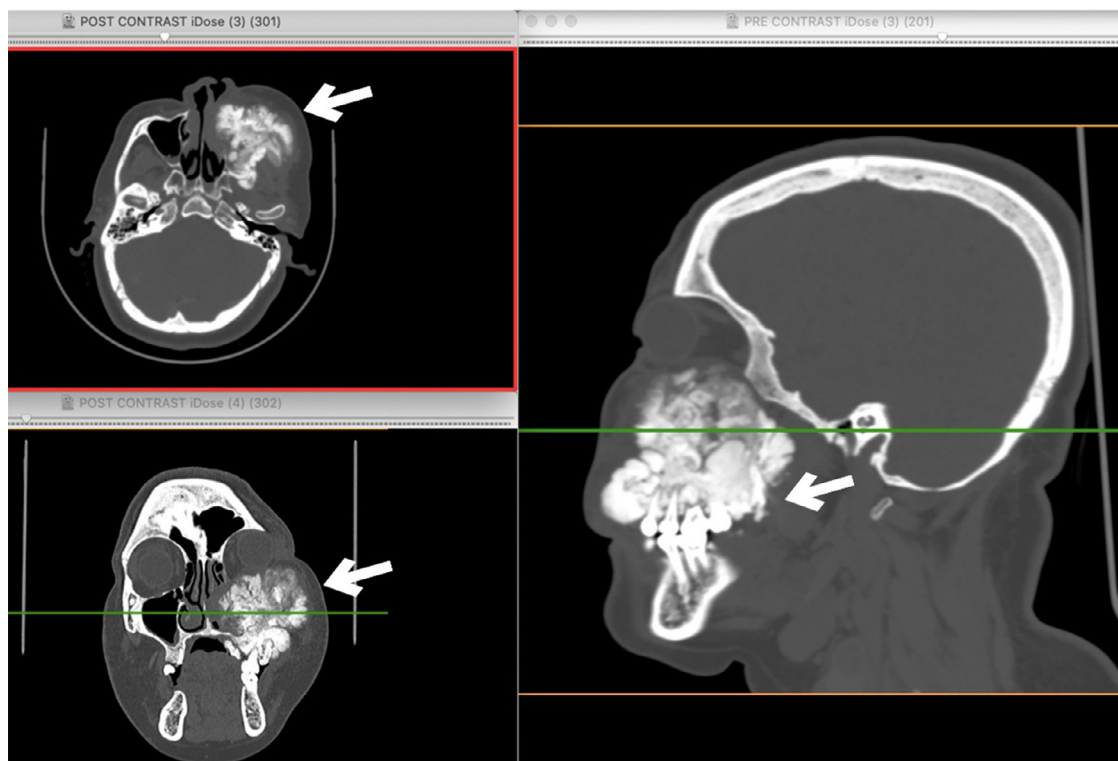


Fig. 3 – Computed tomography (CT) scan showing chondrogenic type osteosarcoma.

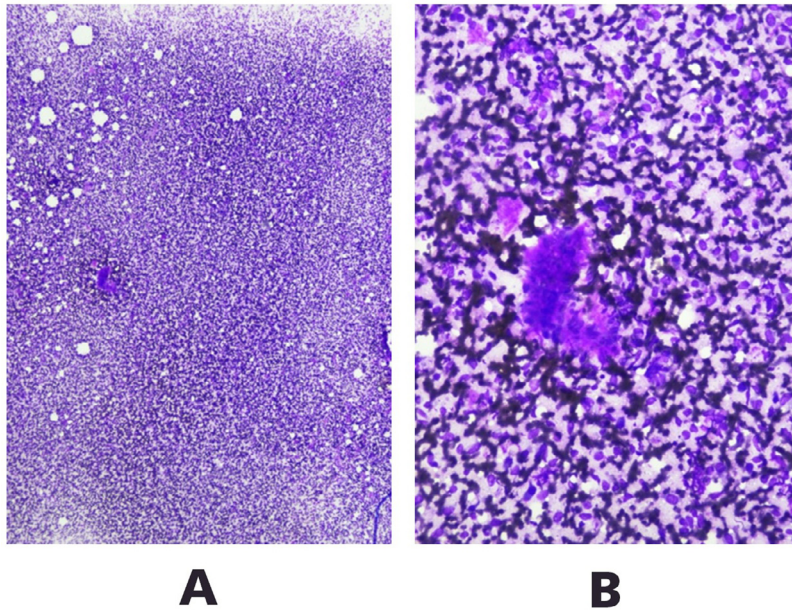


Fig. 5 – Fine-needle aspiration biopsy (FNAB) evaluation with diff-quick stained tissue (A. original magnification 20x) and (B. original magnification 10x) showing hypercellular specimens with anaplastic cells with pleomorphic oval-round nuclei. Some cells are binucleated and hyperchromatic, with decent cytoplasm and eosinophilic matrix.

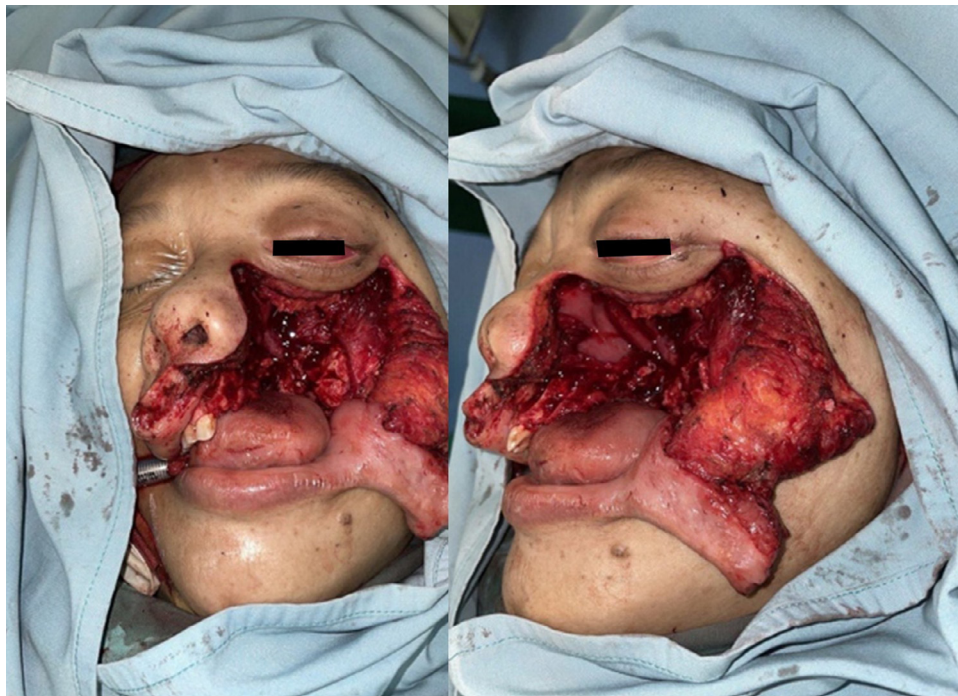


Fig. 6 – Total maxillectomy.

factors for OS are poorly understood, but some predisposing factors include trauma, extrinsic chemicals and carcinogens, and viruses. In our case this occurred in a 48-year-old female with a mass located in the maxilla which is very rare. Furthermore, In this case not found any risk factor, but ethnicity can be a contributing factor to the development of OS.

The clinical signs of OS may be present for a significant amount of time, sometimes weeks to months, before patients seek evaluation. Most patients present with symptoms such as bone pain that has been misinterpreted as a sprain, arthritis, or growing pains. A swelling or mass may or may not be reported depending on tumor size and location. Systemic symptoms, such as those seen in lymphoma patients (eg, fever and

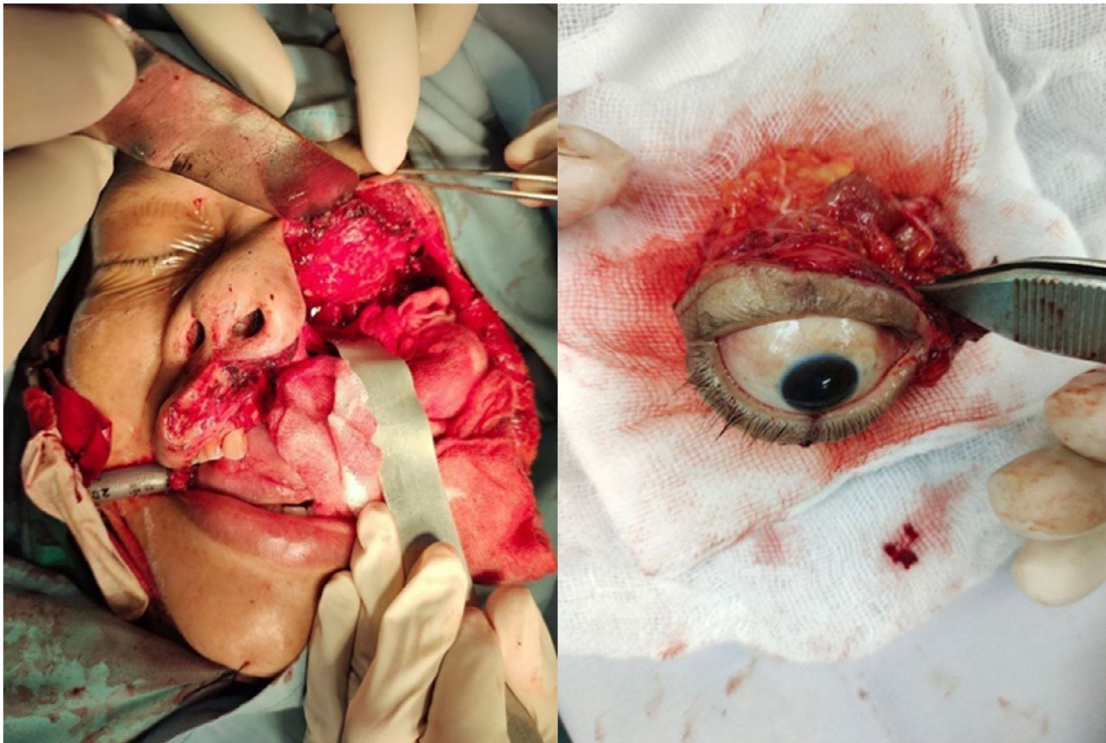


Fig. 7 – Extended exenteration OS.



Fig. 8 – Installation of a surgical obturator, closure, and reconstruction with titanium mesh.

night sweats), are rare. Respiratory symptoms indicate extensive lung involvement. Additional symptoms are unusual because metastasis to other sites is extremely rare. Physical examination findings are focused around the site of the primary tumor. They consist of a palpable mass that may be tender and warm, with or without overlying pulsation or bruit; however, these signs are nonspecific. Joint involvement with decreased range of motion, local or regional lymphadenopathy (unusual), and respiratory findings with metastatic involvement can also be observed [5]. In our case, patient presented with swelling of the left cheek accompanied by palpable solid mass with pain growing in size in the last 3 years and there are no any additional symptoms that lead to metastasis to other site. There are symptoms suggestive of complications presented with left epiphora, exophthalmos, blurry vision; and rhinitis. A physical examination revealed a single mass 6 cm × 5 cm × 10 cm in the left maxillary-nasal-infraorbital region, fixed, hard-dense consistency, irregular surface, indistinct boundaries, and pain (Fig. 1).

Examination begins with orthogonal radiographs of the affected extremity. Radiographs will typically demonstrate a poorly margined or moth-eaten appearance of the bone with mixed amounts of cloudy mineralized matrix and areas of bone resorption. Alternatively, a cartilage or fibrous matrix may be present, or there may be tremendous bone resorption, depending on the subtype. Lab work is nondiagnostic, but high levels of alkaline phosphatase and lactate dehydrogenase have been shown to predict a poorer prognosis. CT scan delineates the bony anatomy like cortical integrity more clearly and picks up pathology fracture and is helpful

in assessing ossification and calcification (condroid component) more accurately. MRI will clearly demonstrate the extent of the bone marrow invasion, the presence and size of any soft-tissue mass, and the relationship to surrounding vital structures. Generally speaking, CT is inferior to MRI, unless further information is needed regarding cortical integrity or the presence of fracture [6,7]. When a diagnosis of malignancy is suspected, a biopsy is required for tissue confirmation. This can usually be accomplished with a core needle biopsy using either ultrasound or CT guidance [6,8]. Our patient showed permeative type maxillary sinus destruction with a sunburst periosteal reaction, a broad transitional zone, and chondroid type matrix on panoramic radiographs (Fig. 2). CT findings suggested the presence of an aggressive primary malignant bone tumor (most likely chondrogenic type OS); right and left maxillary sinusitis; right submandibular lymphadenopathy; and subcentimetric left submandibular, submental, and bilateral posterior trigonum lymph nodes (Fig. 3). No pulmonary metastases were observed on thoracic radiographs (Fig. 4). FNAB revealed hypercellular specimens with anaplastic cells with oval-round nuclei. Some cells were binucleated and hyperchromatic, with decent cytoplasm and eosinophilic matrix (Fig. 5). In our case, we did not carry out further MRI examinations because of the modalities that we had carried out and discussions with the multidisciplinary team, we were able to decide to immediately carry out therapy for this patient.

Contemporary therapy focuses on local and systemic manifestations of disease through the use of multidisciplinary therapy, surgery, and chemotherapy [9]. In our case, a total maxillectomy was performed; the orbital floor, medial wall of the orbit, and ethmoid sinus were also resected (Fig. 6) and extended exenteration OS (Fig. 7). Finally, we installed a surgical obturator and closed and reconstructed the wound with titanium mesh (Fig. 8). Fortunately, our patient did not experience any postsurgical complications and was discharged after 5 days of treatment.

Conclusion

We report the case of a 48-year-old female who presented with swelling of the left cheek. A CT scan suggested the presence of an aggressive primary malignant bone tumor (most likely chondrogenic type OS); right and left maxillary sinusitis; right submandibular lymphadenopathy; and subcentimetric left submandibular, submental, and bilateral posterior trigonum lymph nodes. A biopsy obtained at the time of surgery confirmed a diagnosis of mixed type maxillary OS.

Ethical approval

This study met all ethical principles and received approval from the Research Ethics Committee of the Department of Radiology, Faculty of Medicine, Universitas Airlangga, Dr Soetomo Academic General Hospital, Surabaya, Indonesia.

Patient consent

Written informed consent was obtained from the patient for the publication of this case report, as there are no identifiable patient data included in this report.

Acknowledgments

None.

REFERENCES

- [1] Lindsey BA, Markel JE, Kleinerman ES. Osteosarcoma overview. *Rheumatol Ther*. 2017;4(1):25–43. doi:10.1007/s40744-016-0050-2.
- [2] Amirhossein M, et al. Osteosarcoma: a comprehensive review. *SICOT J*. 2018;4:12. doi:10.1051/sicotj/2017028.
- [3] Bielack S, Kempf-Bielack B, Delling G, et al. Prognostic factors in high-grade osteosarcoma of the extremities or trunk: An analysis of 1702 patients treated on neoadjuvant Cooperative Osteosarcoma Study Group protocols. *J Clin Oncol* 2002;20:776–90.
- [4] Ottaviani G, Jaffe N. The epidemiology of osteosarcoma. *Cancer Treat Res* 2009;152:3–13. doi:10.1007/978-1-4419-0284-9_1.
- [5] Widhe B, Widhe T. Initial symptoms and clinical features in osteosarcoma and Ewing sarcoma. *J Bone Joint Surg Am* 2000;82(5):667–74. doi:10.2106/00004623-200005000-00007.
- [6] Durfee RA, Mohammed M, Luu HH. Review of osteosarcoma and current management. *Rheumatol Ther* 2016;3(2):221–43. doi:10.1007/s40744-016-0046-y.
- [7] Fox MG, Trotta BM. Osteosarcoma: review of the various types with emphasis on recent advancements in imaging. *Semin Musculoskelet Radiol* 2013;17(2):123–36. doi:10.1055/s-0033-1342969.
- [8] Clayer M. Many faces of osteosarcoma on plain radiographs. *ANZ J Surg* 2015;85(1–2):22–6. doi:10.1111/ans.12836.
- [9] Fletcher CDM, Unni K, Mertens F, editors. World Health Organization Classification of Tumours. Pathology and Genetics of Tumours of Soft Tissue and Bone. IARC Press; 2002 <http://www.iarc.fr/WHO-BlueBooks/BBwebsite/bb5.html>.